```
// Name: Utkarsha Patil
// Roll no: SEA52
// Graph Traversal for Campus Navigation using BFS and DFS

#include <stdio.h>
void BFS(int vrtx, int arr[vrtx][vrtx], int start){
   int visited[vrtx];
   int queue[vrtx];
   int front = 0, rear = -1;
   for(int i = 0; i < vrtx; i++){
      visited[i] = 0;
   }
   visited[start] = 1;
   queue[++rear] = start;
   printf("BFS Traversal: ");
   while(front <= rear){
      int node = queue[front++];
      printf("%d ", node);
      for(int i = 0; i < vrtx; i++){
         if(arr[node][i] == 1 && !visited[i]){
            visited[i] = 1;
            queue[++rear] = i;
         } } }
   printf("\n");
}
void DFS(int vrtx, int arr[vrtx][vrtx], int visited[], int node){
   visited[node] = 1;
   printf("%d ", node);
```

```
    for(int i = 0; i < vrtx; i++){
        if(arr[node][i] == 1 && !visited[i]){
            DFS(vrtx, arr, visited, i);
        } } }
void main(){
    int vrtx, edg;
    printf("Enter Number of Buildings: ");
    scanf("%d", &vrtx);
    printf("Enter Number of Paths: ");
    scanf("%d", &edg);
    int arr[vrtx][vrtx];
    int x, y;
    for(int i = 0; i < vrtx; i++){
        for(int j = 0; j < vrtx; j++){
            arr[i][j] = 0;
        } }
    printf("\n--//PATHS//--\n");
    for(int i = 1; i <= edg; i++){
        printf("Enter Path %d (vertex numbers 0 to %d): ", i, vrtx - 1);
        scanf("%d %d", &x, &y);
        arr[x][y] = arr[y][x] = 1;
    }
    printf("\nAdjacency Matrix for given data:\n");
    for(int i = 0; i < vrtx; i++){
        for(int j = 0; j < vrtx; j++){
            printf("%d ", arr[i][j]);
        }
```

```
        printf("\n");
    }
    int start;
    printf("\nEnter Starting Building (0 to %d): ", vrtx - 1);
    scanf("%d", &start);
    BFS(vrtx, arr, start);
    int visited[vrtx];
    for(int i = 0; i < vrtx; i++){
        visited[i] = 0;
    }
    printf("DFS Traversal: ");
    DFS(vrtx, arr, visited, start);
    printf("\n");
}

// Output
Enter Number of Buildings: 5
Enter Number of Paths: 6

--//PATHS//--
Enter Path 1 (vertex numbers 0 to 4): 0 1
Enter Path 2 (vertex numbers 0 to 4): 0 2
Enter Path 3 (vertex numbers 0 to 4): 0 3
Enter Path 4 (vertex numbers 0 to 4): 1 4
Enter Path 5 (vertex numbers 0 to 4): 2 4
Enter Path 6 (vertex numbers 0 to 4): 3 4

Adjacency Matrix for given data:
```

0 1 1 1 0

1 0 0 0 1

1 0 0 0 1

1 0 0 0 1

0 1 1 1 0

Enter Starting Building (0 to 4): 0

BFS Traversal: 0 1 2 3 4

DFS Traversal: 0 1 4 2 3